ChE 382: Unit Operations Laboratory

FLUID FLOW THROUGH FLUIDIZED BEDS

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Introduction

Fluidized beds contain small particles of solids through which a fluid flows with sufficient velocity such that the particles are in motion. If a fluid flows slowly upward through a bed of fine particles at a low velocity, the fluid simply passes through the void spaces, and the particles remain stationary. As the fluid flow rate is increased, the particles tend to move apart slightly resulting in an observable increase in the bed height, and in some regions of the bed, particles may begin to show slight movement. As still higher fluid velocities, the particles in the bed are brought into a rapid motion and circulate freely in the bed; the bed height is ill-defined but particles are not yet carried out with the gas. At this point, the bed of particles is fluidized. At fluidization conditions, a large fraction of the gas flow passes through the bed in the form of "bubbles." These bubbles are (surprisingly) 'stiff,' and they provide the driving force for the mixing and circulation of the solids.

Fluidized beds are typically used in reaction engineering applications. The solid particles are usually a catalyst (or a mixture of catalyst and inert particles) or solids such as coal or an ore that are directly combusted or roasted. The particles in a fluidized bed are typically quite small, and a range of 50-500 μm is not uncommon.

Because of the rapid mixing of particles that occurs in a fluidized bed, the solid phase may be modeled as being well-mixed and thus of uniform temperature and composition. The gas flow in the reactor, though, is quite complex; a portion of the gas passes through the bed of solids while the remainder tends to move through the bed in the form of bubbles in a way similar to a gas passing upward through a liquid.

In the experiment, we will study the fluidization behavior of a bed of two types of small particles, and measure the gas pressure drop across the bed of particles as a function of the gas velocity. The dependence of the pressure drop on the superficial gas velocity is shown in Figure 1 (at end of lab manual) and in the Unit Operation textbook, Chapter 7 Figure 7.11 [1]. At low velocities, the pressure drop is that of gas flow through a packed bed and can be determined from the Ergun equation. At point A in FIGURE 1, the pressure drop reaches a maximum. With a slight additional increase in gas velocity, the bed unlocks resulting in a small decrease in the pressure drop and a small increase in the bed voidage from its static value. This is now an incipiently fluidized bed. Further increases in gas velocity beyond the incipient fluidization velocity do not result in any appreciable increase in the pressure drop, although the movement and mixing of the solids becomes increasingly intense. The additional gas flow passes through the bed in the form of bubbles, which move through the deformable bed of solids without any significant hydrodynamic resistance. With decreasing gas velocity, the pressure drop curve does not follow exactly the curve for increasing pressure. There is a significant hysteresis, because with increasing flow, the system must overcome the coefficient of friction for the static bed to set the particles in motion.

Theory – Fluidized Beds

Pressure Drop

At low gas velocities, the pressure drop through the bed can be described by the Ergun (1) equation. (Eqn. 7.22 in Ref. [1]):

$$\frac{\Delta P}{L} = \frac{150\bar{V}_0\mu}{\Phi_s^2 D_p^2}\frac{(1-\varepsilon)^2}{\varepsilon^3} + \frac{1.75\rho\bar{V}_0^2}{\Phi_s D_p}\frac{1-\varepsilon}{\varepsilon^3} \tag{1}$$

Where:
ΔP is the pressure drop across the bed
L is the bed height
ρ is the fluid density
$\bar{V}_0$ is the superficial velocity (velocity based on the empty cross section of the bed)
ε is the bed voidage at minimum fluidization
μ is the fluid viscosity
Φ_s is the sphericity (see table 1 in Ref. [1])
D_p is the particle diameter (see Ref. [1] p.164)

Minimum Velocity of Fluidization

The fluid mechanics in a fluidized bed are very complex. Hence, chemical engineers have developed semi-empirical correlations for the minimum velocity of fluidization based upon the particle Reynolds number. One source for these equations is McCabe and Smith [1]. More detailed presentations are given by Davidson and Harrison [2], Kunii and Levenspiel [3], and Perry's Chemical Engineering Handbook [4]. The basis of these correlations is that the minimum fluidization velocity, $\bar{V}_{0M}$ for the uniform-sized particles can be predicted by using the force balance that at minimum fluidization the drag force is exerted on the bed of particles by the fluid is equal to the force of gravity on the bed. Further details on the equation used for the minimum fluidization velocity are given in Chapter 7, pp 177-189 (Unit Operations textbook) [1].

Experimental

Before starting the experiment, familiarize yourself with the lab setup. The fluidized bed vessel is constructed of a 4" plexiglass tube with an enlargement at the top of the tube to reduce entrainment losses (left vessel will be used). Air is introduced at the bottom of the column through a porous metal disc, which serves as the bed support. Air flow is controlled by adjusting valves located upstream of two rotameters. The pressure drop across the bed is measured by a manometer.

Experimental Procedure

Two sets of experiments will be performed: one with a mixture of fine silica sand and the other with ion exchange resin beads as the bed material.

Sand and Ion-Exchange Resin Particle Size (D_p)Measurements

The particle size D_{pav}, is needed in the minimum fluidization velocity equation. The sand consists of a distribution of particle sizes. Use a set of sieves to determine the size distribution of the sand. The average particle size for each batch of sand is calculated as follows:

$$D_{p_{av}} = \frac{1}{\sum\{x_i/D_{p_i}\}}$$

where x_i is the mass fraction of particles of size D_{p_i}. The average particle diameter, $D_{p_{av}}$, should be used in evaluating equations in this section. Also plot the mass vs particle size to determine the particle size distribution. Refer to Chapter 28, "Properties and Handling of Particulate Solids", in McCabe , Smith and Harriott for more details on how to conduct a screen analysis [1].

The particle size of the ion-exchange resin can also be measured using the sieves.

Fluidization Experiments

Charge the vessel with the first bed material to a packed height of 6". Slowly turn on the air and record the pressure drop using the manometer as a function of the gas flowrate. Continue to increase the gas flow until conditions beyond minimum fluidization are reached. This can be judged by observing that the pressure drop does not increase significantly with greater gas flowrates. Next decrease the gas velocity and record the pressure drop across the bed as the gas velocity is decreased to zero. Measurements should be made for static bed heights of 6 and 10 inches with both bed materials.

DATA ANALYSIS

Plot the measured pressure drop per unit bed height as a function of the gas velocity (or flowrate) for both increasing and decreasing flowrates. Compare your measured $\bar{V}_{0M}$ to the value calculated by equation 7.51 in Reference [1]. Note that you need the bed porosity, ε_M, to use equation 7.51. This may be measured experimentally, or values of porosity may be found in a handbook. Below $\bar{V}_{0M}$, the pressure vs. flowrate data can be predicted by the Ergun equation (1). How well do your measured data agree with the predicted values in this regime?

References

[1] W. McCabe, J. Smith, and P. Harriot, *Unit Operations of Chemical Engineering,* 7th ed. McGraw-Hill, 2005.

[2] J. Davidson, R. Clift, and D. Harrison, Eds., *Fluidization*, 2nd ed. Academic Press, 1995

[3] D. Kunii and O. Levenspiel, *Fluidization Engineering*, 2nd ed. Elsevier Science and Technology, 1992.

[4] R. H. Perry and D. W. Green, Eds., *Perry's Chemical Engineer's Handbook,* 7th ed. McGraw-Hill, 1997.

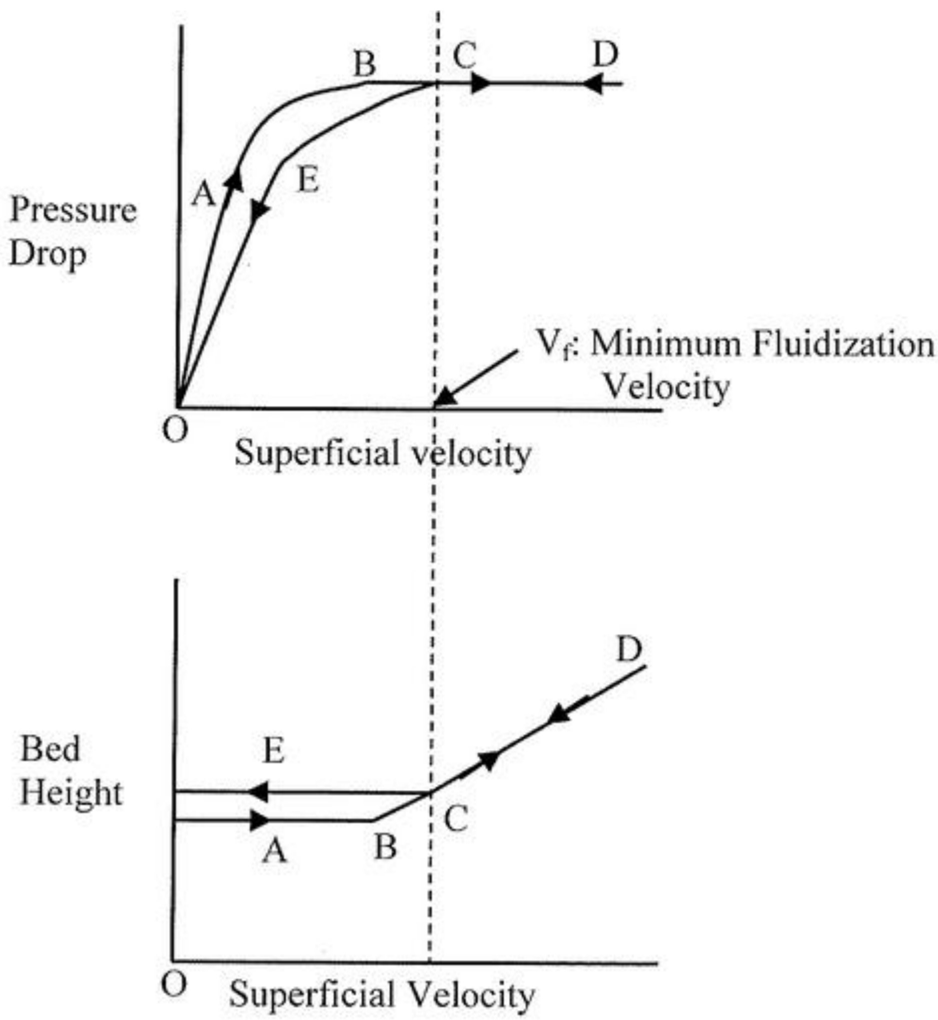

FIGURE 1. Pressure Drop and bed porosity or superficial air velocity. Arrows refer to increasing and decreasing gas flows